

Data Cleaning Report

Aspect Based Sentiment Analysis for Laptop Reviews

1. Data Source

The dataset used in this project is a collection of laptop reviews scraped from Flipkart, provided as a CSV file named `laptops_dataset_final_600.csv`. The raw dataset contains 24,113 individual reviews across 365 distinct laptop products. Each row includes the product name, the product's overall rating, the total number of ratings and reviews the product has received, the individual reviewer's star rating, a short review title, and the full review text.

2. Data Quality Issues Identified

During the data understanding stage, the following data quality issues were identified in the raw dataset:

- Duplicate rows. The dataset contained 7,122 exact duplicate rows, roughly 30 percent of all rows. Investigation showed these were long, uniquely worded reviews repeated in full under the same product, with identical text, rating, and formatting. This pattern is highly unlikely to occur from different customers independently, and points instead to the review being captured more than once during scraping.
- Incorrect data types. The columns `no_ratings` and `no_reviews` were stored as text instead of numbers. Most individual values looked like plain numbers, for example 900, but a small number of values used commas as thousands separators, for example 9,156 or 15,210. Since pandas assigns one data type per column based on every value in it, the presence of even a few comma separated values forced both columns to be stored as text across all rows.
- Skewed rating distribution. Although not a data error, the star rating distribution was found to be heavily skewed toward positive reviews, with 58 percent of all reviews rated 5 stars and only around 10 percent rated 1 or 2 stars. This is documented as a limitation of the dataset rather than something corrected during cleaning, since it reflects genuine customer behaviour.

3. Cleaning Methods Used

- Removed exact duplicate rows using `pandas drop_duplicates`, which reduced the dataset from 24,113 rows to 16,991 rows.
- Cleaned the `no_ratings` and `no_reviews` columns by removing comma characters from every value using `string replace`, then converting both columns to integer type using `astype`.
- Verified the result by checking that duplicate count was zero, missing value count was zero across all columns, and both cleaned columns showed integer data type, before saving the cleaned file.

4. Justification for Decisions

The decision to remove all 7,122 duplicate rows, rather than only reviewing them individually, was based on the fact that pandas `drop_duplicates` only removes rows that are identical across every single column at once, meaning the product, rating, title, and full review text all match exactly. This makes it very unlikely that any genuine, independently written customer opinion was removed. Short generic reviews such as a single word were also examined, and even these were only removed when every column matched exactly, which keeps the same safety guarantee.

The decision to convert `no_ratings` and `no_reviews` to numeric type, rather than leave them as text, was necessary because these columns represent counts and are needed for any future numeric analysis or filtering, for example identifying the most reviewed products. Leaving them as text would silently break any such calculation without raising an error, which is a more dangerous failure mode than fixing it at the cleaning stage.

5. Features Engineered

No new features were engineered during this cleaning stage. Feature engineering specific to aspect based sentiment analysis, such as extracting mentioned laptop features and sentiment scores per review, is planned for a later stage of the project once the cleaned dataset is used for aspect extraction.

6. Challenges Faced

The main challenge during this stage was not in the data itself but in the git workflow used to manage the project. While setting up branch based version control, the local main branch and the remote main branch on GitHub were found to have unrelated commit histories, caused by an automatically generated README file created when the GitHub repository was first set up. This was resolved by confirming the conflicting commit contained no important content, then synchronising the remote branch with the local project history. A secondary issue involving a locked git index file on Windows, caused by an interrupted operation, was also resolved by manually clearing the lock file. Both issues were investigated before any action was taken, to avoid the risk of losing project work.

7. Conclusion

Following the cleaning steps described above, the dataset was reduced from 24,113 rows to 16,991 rows, with all duplicate rows removed, correct numeric data types applied to `no_ratings` and `no_reviews`, and zero missing values across all columns. The cleaned dataset has been saved as `data/processed/processed_data.csv` and is ready to be used in the next stage of the project, aspect extraction.